2nd Midterm · Unit 6 · Laplace equation by separation · 5 points · 45 min · Standard exam level

P13 — Laplace's equation in a rectangle with Neumann conditions on the vertical sides and Dirichlet data given by finite cosine sums: the zero mode, cosh and sinh modes, flux balance and uniqueness

The steady temperature $u(x,y)$ of the plate $0<x<\pi$, $0<y<1$ satisfies Laplace's equation

$$u_{xx}+u_{yy}=0.$$

The vertical sides are insulated and the horizontal sides are held at prescribed temperatures:

$$u_x(0,y)=u_x(\pi,y)=0, \qquad u(x,0)=2+\cos 2x, \qquad u(x,1)=4+3\cos x.$$

(a) Separate variables, $u=X(x)Y(y)$, and solve the eigenvalue problem for X, discussing the three signs of the separation constant.

(b) Solve the equations for Y, paying special attention to the zero eigenvalue, and write the general solution that satisfies the conditions on the insulated sides.

(c) Impose the conditions on $y=0$ and $y=1$ and obtain $u(x,y)$ in closed form. Give u at the centre $(\pi/2,\ 1/2)$.

(d) Compute $\Phi(y)=\displaystyle\int_0^{\pi} u_y(x,y)\,dx$ and explain why it does not depend on y. Is the solution unique? Would it be unique, and would it exist for any data, if the horizontal sides were also insulated?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,y)$ on the rectangle $R=(0,\pi)\times(0,1)$; steady two-dimensional conduction, no sources.
- Operator: Laplacian, linear, second order, elliptic, homogeneous equation. The conditions are homogeneous (Neumann) in x and non-homogeneous (Dirichlet) in y, so the eigenvalue problem is posed in x.
- Data: the bottom and top temperatures are **finite cosine sums**, i.e. already combinations of the eigenfunctions found in (a); no integral is needed for the coefficients.

Phase 2. Canonical formulation

$$\Delta u = 0 \text{ in } R, \qquad \frac{\partial u}{\partial x} = 0 \text{ on } x = 0, \pi, \qquad u = f(x) \text{ on } y = 0, \qquad u = g(x) \text{ on } y = 1$$

with $f(x) = 2 + \cos 2x$ and $g(x) = 4 + 3\cos x$. Sturm-Liouville problem in x: $X'' + \lambda X = 0, X'(0) = X'(\pi) = 0$.

Part (a) · 1.5 points

Separation and the eigenvalue problem in x

Step 1. Separation

$u = X(x)Y(y)$ gives $X''Y + XY'' = 0$. Dividing by XY: $\dfrac{X''}{X} = -\dfrac{Y''}{Y}$. The left side depends only on x, the right only on y; both equal a constant $-\lambda$:

$$X'' + \lambda X = 0, \qquad Y'' - \lambda Y = 0$$

The side conditions $u_x(0, y) = X'(0)Y(y) = 0$ for all y and $Y \not\equiv 0$ give $X'(0) = 0$; likewise $X'(\pi) = 0$.

Step 2. Case lambda < 0

$\lambda = -m^2, m > 0$: $X = a\cosh mx + b\sinh mx$, $X' = m(a\sinh mx + b\cosh mx)$. $X'(0) = mb = 0 \Rightarrow b = 0$. $X'(\pi) = ma\sinh m\pi = 0$ with $\sinh m\pi > 0 \Rightarrow a = 0$. Trivial.

Step 3. Case lambda = 0

$X'' = 0, X = a + bx, X' = b$. Both conditions give $b = 0$, and a is free: $X_0 = 1$ is an eigenfunction. With Neumann conditions the zero eigenvalue **does** occur (Haberman Sec. 2.4).

Step 4. Case lambda > 0

$\lambda = m^2$: $X = a\cos mx + b\sin mx$, $X' = m(-a\sin mx + b\cos mx)$. $X'(0) = mb = 0 \Rightarrow b = 0$.
$X'(\pi) = -ma\sin m\pi = 0$ with $a \neq 0$ requires $m = n \in \{1, 2, \dots\}$. So $\lambda_n = n^2$, $X_n = \cos nx$.

KEY RESULTS

- $\dfrac{X''}{X} = -\dfrac{Y''}{Y} = -\lambda$; $X'' + \lambda X = 0, X'(0) = X'(\pi) = 0$; $Y'' - \lambda Y = 0$.
- $\lambda < 0$: only $X \equiv 0$.
- $\lambda_0 = 0$ is an eigenvalue with $X_0 = 1$.
- $\lambda_n = n^2, X_n = \cos nx, n = 1, 2, \dots$

Part (b) · 1 point

The y equations and the general solution

Step 1. Zero mode

$\lambda = 0$: $Y'' = 0$, integrate twice: $Y' = B_0$, $Y = A_0 + B_0 y$. The product $X_0 Y_0 = A_0 + B_0 y$ is the part of the temperature that varies linearly between the plates.

Step 2. Modes n >= 1

$Y'' - n^2 Y = 0$ has characteristic equation $r^2 - n^2 = 0$, $r = \pm n$, so $Y = \alpha e^{ny} + \beta e^{-ny}$. With $\cosh ny = \frac{1}{2}(e^{ny} + e^{-ny})$ and $\sinh ny = \frac{1}{2}(e^{ny} - e^{-ny})$ this is $Y_n = A_n \cosh ny + B_n \sinh ny$ ($A_n = \alpha + \beta$, $B_n = \alpha - \beta$). The cosh / sinh form is preferred because $\sinh 0 = 0$ and $\cosh 0 = 1$ make the condition at $y = 0$ immediate.

Step 3. Superposition

Every product satisfies Laplace's equation and the side conditions, and so does the sum:

$$u(x, y) = A_0 + B_0 y + \sum_{n=1}^{\infty} \cos nx \left(A_n \cosh ny + B_n \sinh ny \right)$$

KEY RESULTS

- $Y_0 = A_0 + B_0 y$ (linear in y).
- $Y_n = A_n \cosh ny + B_n \sinh ny$.
- $u(x, y) = A_0 + B_0 y + \sum_{n=1}^{\infty} \cos nx \left(A_n \cosh ny + B_n \sinh ny \right)$.

Part (c) · 1.5 points

Coefficients from the data and the closed form

Step 1. Bottom side

$u(x, 0) = A_0 + \sum_n A_n \cos nx = 2 + \cos 2x$. By orthogonality of $\{1, \cos nx\}$ on $(0, \pi)$ the coefficients are unique: $A_0 = 2$, $A_2 = 1$, $A_n = 0$ for $n \neq 2$. (Directly: $\frac{1}{\pi} \int_0^{\pi} (2 + \cos 2x) dx = 2$ and $\frac{2}{\pi} \int_0^{\pi} (2 + \cos 2x) \cos nx \, dx = \delta_{n2}$.)

Step 2. Top side

$u(x,1) = A_0 + B_0 + \sum_n (A_n \cosh n + B_n \sinh n)\cos nx = 4 + 3\cos x$:

- constant: $2 + B_0 = 4 \Rightarrow B_0 = 2$;
- $n = 1$: $0 \cdot \cosh 1 + B_1 \sinh 1 = 3 \Rightarrow B_1 = \dfrac{3}{\sinh 1}$;
- $n = 2$: $1 \cdot \cosh 2 + B_2 \sinh 2 = 0 \Rightarrow B_2 = -\dfrac{\cosh 2}{\sinh 2} = -\coth 2$;
- $n \geq 3$: $B_n \sinh n = 0 \Rightarrow B_n = 0$.

Step 3. Simplify the n = 2 mode

$$\cosh 2y - \frac{\cosh 2}{\sinh 2}\sinh 2y = \frac{\sinh 2\cosh 2y - \cosh 2 \sinh 2y}{\sinh 2} = \frac{\sinh(2-2y)}{\sinh 2}$$

by $\sinh(a-b) = \sinh a \cosh b - \cosh a \sinh b$. It equals 1 at $y = 0$ and 0 at $y = 1$, as required. Therefore

$$u(x,y) = 2 + 2y + \frac{3\sinh y}{\sinh 1}\cos x + \frac{\sinh\big(2(1-y)\big)}{\sinh 2}\cos 2x$$

Step 4. Check

$u_{xx} = -\dfrac{3\sinh y}{\sinh 1}\cos x - \dfrac{4\sinh(2-2y)}{\sinh 2}\cos 2x$ and $u_{yy} = \dfrac{3\sinh y}{\sinh 1}\cos x + \dfrac{(-2)^2\sinh(2-2y)}{\sinh 2}\cos 2x$ (chain rule: $\frac{d}{dy}\sinh(2-2y) = -2\cosh(2-2y)$), so $u_{xx} + u_{yy} = 0$. u_x contains $\sin x$ and $\sin 2x$, both zero at $x = 0, \pi$.

Step 5. Centre

$\cos(\pi/2) = 0$, $\cos\pi = -1$: $u(\pi/2, 1/2) = 2 + 1 - \dfrac{\sinh 1}{\sinh 2}$. With $\sinh 2 = 2\sinh 1 \cosh 1$:

$$u\Big(\frac{\pi}{2}, \frac{1}{2}\Big) = 3 - \frac{1}{2\cosh 1} \approx 3 - 0.324027 = 2.675973$$

KEY RESULTS

- $A_0 = 2, B_0 = 2$; $A_1 = 0, B_1 = \dfrac{3}{\sinh 1}$; $A_2 = 1, B_2 = -\coth 2$; all other coefficients 0.
- $u(x,y) = 2 + 2y + \dfrac{3\sinh y}{\sinh 1}\cos x + \dfrac{\sinh\big(2(1-y)\big)}{\sinh 2}\cos 2x$.
- $u(\pi/2, 1/2) = 3 - \dfrac{1}{2\cosh 1} \approx 2.675973$.

Part (d) · 1 point

Flux balance and uniqueness

Step 1. The integral

$u_y = 2 + \dfrac{3\cosh y}{\sinh 1}\cos x - \dfrac{2\cosh(2-2y)}{\sinh 2}\cos 2x$. Since $\int_0^\pi \cos x\,dx = [\sin x]_0^\pi = 0$ and $\int_0^\pi \cos 2x\,dx = \left[\tfrac{1}{2}\sin 2x\right]_0^\pi = 0$:

$$\Phi(y) = \int_0^\pi 2\,dx = 2\pi$$

Step 2. Why it is constant

$\Phi'(y) = \displaystyle\int_0^\pi u_{yy}\,dx = -\int_0^\pi u_{xx}\,dx = -\big[u_x\big]_{x=0}^{x=\pi} = -(0-0) = 0$. Physically the heat flux $-K\,u_y$ through any horizontal line is the same: there are no sources inside and the sides are insulated. The temperature rises linearly on average, from mean 2 at the bottom to mean 4 at the top, so $B_0 = 2$ is exactly the mean gradient.

Step 3. Uniqueness (Dirichlet on two sides)

Let u_1, u_2 be two solutions and $w = u_1 - u_2$: $\Delta w = 0$, $w_x = 0$ on $x = 0, \pi$, $w = 0$ on $y = 0, 1$. Green's first identity (extension: $\iint_R \left(w\Delta w + |\nabla w|^2\right)dA = \oint_{\partial R} w\,\dfrac{\partial w}{\partial n}\,ds$) gives $\iint_R |\nabla w|^2 dA = 0$, because on each side either $w = 0$ or $\partial w/\partial n = 0$. Hence $\nabla w = 0$, w is constant, and $w = 0$ on $y = 0$ makes it zero. The solution is unique. Consistently, the maximum principle (Haberman Sec. 2.5) bounds u between the extreme boundary values: $1 = u(\pi/2, 0) \le u \le u(0, 1) = 7$ (the maximum principle on this mixed problem is checked on a grid in the script).

Step 4. All sides insulated

If u_y were prescribed on $y = 0$ and $y = 1$ as well, the zero mode would only fix B_0 (twice) and leave A_0 free: solutions would differ by constants. Moreover, by Step 2 the prescribed fluxes would have to satisfy $\int_0^\pi u_y(x, 0)\,dx = \int_0^\pi u_y(x, 1)\,dx$ (solvability condition, Haberman Sec. 2.5): no net heat may enter a steady insulated plate. Data violating it give no solution at all.

KEY RESULTS

- $\Phi(y) = 2\pi$ for every y: only the zero mode carries net flux.
- Constant because, integrating $u_{xx} + u_{yy} = 0$ over $(0, \pi)$, $\Phi'(y) = -[u_x]_0^\pi = 0$: what enters through one plate leaves through the other (insulated sides).
- Unique: with Dirichlet data on two sides, $w \equiv 0$. With all four sides insulated, solutions exist only if the net flux data are compatible, and are unique only up to an additive constant.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** the $\cos 2x$ mode imposed at the bottom decays upward like $\sinh(2(1-y))/\sinh 2 \approx e^{-2y}$; higher modes $\cos nx$ would decay like e^{-ny}, so fine detail of boundary data is felt only near the side where it is imposed. The zero mode does not decay: it is the only part of the data felt across the whole plate.
- **Checks:** $u(x,0) = 2 + 0 + 0 + \cos 2x$ and $u(x,1) = 4 + 3\cos x + 0$; $u_x(0,y) = u_x(\pi,y) = 0$; the centre value 2.676 lies between the boundary extremes 1 and 7.
- **Common mistakes in the exam:** (1) discarding $\lambda = 0$ as "trivial", which loses the linear part $2 + 2y$ and makes the constant data impossible to match; (2) using $\sin nx$ eigenfunctions (those belong to Dirichlet sides); (3) writing $Y_n = A_n e^{ny} + B_n e^{-ny}$ and then solving a 2×2 system with large numbers instead of using $\sinh(n(1-y))$; (4) claiming uniqueness only up to a constant, which is true only when every side is insulated.